

Transport helpers for the recursive product-density lemma

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Abstract

Step 2a of the general- d equal-ratio monomial milestone (Astra #14). Three transport lemmas for Lebesgue integrals on $(0, 1]$, in $[0, \infty]$ -valued form so that no integrability is needed inside the induction: scaling $\int_{(0,1]} F(tz) dz = \frac{1}{t} \int_{(0,t]} F(w) dw$ for $t > 0$ (`lintegral_Ioc_scale`); the triangle swap $\int_{(0,1]} \int_{(0,t]} G(t, w) dw dt = \int_{(0,1]} \int_{[w,1]} G(t, w) dt dw$ (`lintegral_triangle_swap`); and the FTC identity of unit 165 on $[z, 1]$ in $[0, \infty]$ form (`lintegral_log_sub_pow_div`). Zero sorry/axiom.

1 The things formalised

```
theorem image_mul_Ioc (t) (ht : 0 < t) : (fun z :   => t * z) '' Ioc 0 1 = Ioc 0 t
theorem lintegral_Ioc_scale (t) (ht : 0 < t) (F :   → ENNReal) :
  z in Ioc 0 1, F (t * z) = ENNReal.ofReal (1 / t) *   w in Ioc 0 t, F w
theorem lintegral_triangle_swap (G :   →   → ENNReal)
  (hG : Measurable fun q :   ×   => G q.1 q.2) :
  t in Ioc 0 1,   w in Ioc 0 t, G t w =   w in Ioc 0 1,   t in Icc w 1, G t w
theorem lintegral_log_sub_pow_div (z) (hz : 0 < z) (hz1 : z   1) (d :   ) :
  t in Icc z 1, ENNReal.ofReal ((log t - log z) ^ d / t)
  = ENNReal.ofReal ((-log z) ^ (d + 1) / (d + 1))
```

2 Lean notes

- Scaling is the one-dimensional Jacobian lemma with $f = (t \cdot)$, derivative t and image $(0, t]$; the constant $\frac{1}{t}$ is pulled out with `lintegral_const_mul'`.
- The triangle swap writes each inner integral as an integral over $(0, 1]$ against the indicator of $\{w \leq t\}$; `Measure.restrict_restrict` expects the intersection as `s ∩ t` with the inner set first, and the measurable sets must be spelled `Iic/Ici` for the rewrite to match. Then `lintegral_lintegral_swap` on the restricted product measure.
- `intervalIntegral.integral_of_le`, `setLIntegral_congr Ioc_ae_eq_Icc.symm` and `ofReal_integral_eq_lintegral_ofReal` convert the real FTC identity; interval integrability comes from `ContinuousOn.intervalIntegrable` via `.def'` and `uIoc_of_le`.